

Data Science and Big Data (MCS-226) *(Job/Research-based Course)*

[Block-1, Part-2]

By

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- UNIT 1: INTRODUCTION TO DATA SCIENCE
- UNIT 2: PORTABILITY AND STATISTICS FOR DATA SCIENCE
- UNIT 3: DATA PREPARATION FOR ANALYSIS
 - Need for Data Preparation
 - Data preprocessing
 - Selection and Data Extraction
 - Data Curation
 - Data Integration
 - Knowledge Discovery
- UNIT4: DATA VISUALIZATION AND INTERPRETATION

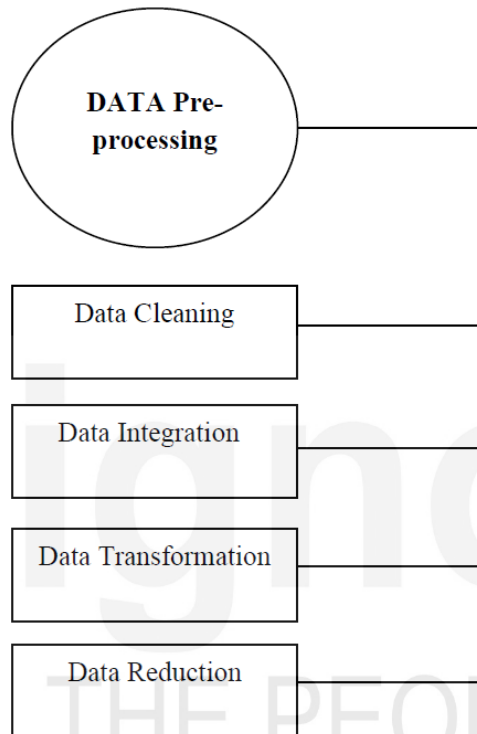
NEED FOR DATA PREPARATION

Data is one of the most significant resources available to marketers, agencies, publishers, media firms, and others today for a reason. But only high-quality data is useful

The basic factors of data quality:

- **Accuracy** - The data must be true and reflect events that actually take place in the real world. Accuracy measures determine how closely the figures agree with the verified right information sources.
- **Completeness** - The degree to which the data is complete determines how well it can provide the necessary values.
- **Consistency** - Data consistency is the homogeneity of the data across applications, networks, and when it comes from several sources. For example, identical datasets should not conflict if they are stored in different locations.
- **Timeliness** - Data that is timely is readily available whenever it is needed. The timeliness factor also entails keeping the data accurate; to make sure it is always available and accessible and updated in real-time.
- **Uniqueness** - Uniqueness is defined as the lack of duplicate or redundant data across all datasets. The collection should contain zero duplicate records.
- **Validity** - Data must be obtained in compliance with the firm's defined business policies and guidelines. The data should adhere to the appropriate, recognized formats, and all dataset values should be within the defined range.

DATA PREPROCESSING



DATA PREPROCESSING:

Data Cleaning	Data Integration	Data Transformation	Data Reduction
a. Missing Values i. Ignore the tuple ii. Manually enter the omitted value iii. Fill up the blank with a global constant iv. To fill in the missing value, use a measure of the attribute's central tendency (such as the mean or median) v. For all samples that belong to the same class as the specified tuple, use the mean or median vi. Fill in the blank with the value that is most likely to be there	a. Entity Identification Problem b. Redundancy and Correlation Analysis c. Tuple Duplication d. Data Value Conflict Detection and Resolution:	a. Dimensionality reduction b. Numerosity reduction c. Transformations d. Data Discretization	a. Smoothing, b. Attribute construction c. Aggregation d. Normalization e. Discretization f. Concept hierarchy creation using nominal data
b. Noisy Data i. Binning ii. Regression iii. Outlier analysis iv. Data discretization	Steps of Data Cleaning: 1. Remove duplicate or irrelevant observations 2. Fix structural errors 3. Managing Unwanted outliers 4. Handling missing data 5. Validate and QA	Methods of Data Cleaning: 1. Ignore the tuples 2. Fill in the missing value 3. Binning method 4. Regression 5. Clustering:	

Steps of Data Transformation...

- **Discretization**: It is the process of transforming continuous variables into discrete variables by creating a set of contiguous intervals, also called bins,
 - that span the range of the variable values.
- Discretization is used to change the distribution of skewed variables and to minimize the influence of outliers, and
 - hence improve the performance of some machine learning models.
- For the variable X, the interval width may be the following binning methods:

1. **Group-wise:**

$$Width = \frac{Max(X) - Min(X)}{Bins}$$

or

2. **Equal-frequency :**

$$Data\ in\ a\ Bin = \frac{Number\ of\ Data}{Bins}$$

For example:

If the values of the variable vary between 0 and 100 and we want 5 Bins, then,

$$Width = (100-0) / 5 = 20$$

Bin 1: 0 - 20

Bin 2: 21 - 40

Bin 3: 41 - 60

Bin 4: 61 - 80

Bin 5: 81 - 100

For example:

Data = 2, 4, 6, 8, 10, 12, 14, 16, 19

Bins = 3, then,

Bin Edge = (1-6], (6-12], (12- 19]

Bin 1: 2, 4, 6

Bin 2: 8, 10, 12

Bin 3: 14, 16, 19

Discretization: Use

- Use1: The first and final bins ([0-20 and 80-100] or Fixed Bin =3) can be expanded to accommodate outliers, that is,
 - values under 0 or greater than 100 would be placed in those bins as well, by extending the limits to minus and plus infinity.
- Use2: Binning uses the “neighborhood” to smooth the storage value of the records. That's the value around it by following methods:
 - i. Smoothing by bin mean
 - ii. Bin borders smoothing

For example:

Data: 2, 4, 6, 8, 10, 12, 14, 16, 19, 20

Bins: 3, then do a Data Discretization.

✓ Bin Edge = (1.9-8], (8-14], (14-20]

Using Equal-frequency:

Bin 1: 2, 4, 6, 8

Bin 2: 10, 12, 14

Bin 3: 16, 19, 20

i) Smoothing by bin mean

Bin 1: 5, 5, 5, 5

Bin 2: 12, 12, 12

Bin 3: 18.3, 18.3, 18.3

- ✓ Each bin values are replaced by the respective bin mean value.

ii) Bin borders smoothing

Bin 1: 2, 2, 8, 8

Bin 2: 10, 10, 14

Bin 3: 16, 19, 20

- ✓ The maximum and minimum values are defined as the boundary values when the bin boundary is smoothed.
- ✓ Each bin value is then replaced by the closest limit value.

- **Normalization:** Data normalization involves converting all data variables into a given range. Some of the techniques that are used for accomplishing normalization are:
 - **Min–max normalization:** This transforms the original data linearly.
 - **Z-score normalization:** In z-score normalization (or zero-mean normalization) the values of an attribute (A), are normalized based on the mean of A and its standard deviation.
 - **Normalization by decimal scaling:** It normalizes the values of an attribute by changing the position of their decimal points

1. Min – Max Normalization

- Which translates the initial data linearly.
- Suppose min X is the minimum and max X is the maximum limit of the variable.
- We’ve got the formula

$$v' = \frac{v - \min X}{\max X - \min X} (\text{new_max}_X - \text{new_min}_X) + \text{new_min}_X$$
- where v is the value that you want to plot in the new set.
- v' is the fresh meaning you get when you normalize the old value.
- Assume the attribute revenue minimum and maximum values are 1000 and 16000 respectively. Diagram income in the [0.0, 1.0] range. For salaries a value of 14000 is converted into a minimum – maximum standardization.

$$= \frac{14000 - 1000}{16000 - 1000} (1.0 - 0) + 0 = 0.866$$

2. Z-Score Normalization

- The value of a variable (X) in the Z-Score Normalization or Zero-Median Normalization is uniform according to the mean of the X and its standard deviation.
- The value v of the X attribute is computer-standardized to v'

$$v' = \frac{v - \bar{X}}{\sigma_X}$$

Where X and σ are the mean and standard deviations respectively of the X attribute.
 This method is helpful if the true minimum and maximum attribute X are not certain or if outliers surpass the minimum – maximum normalization.

- Suppose that the average and standard deviation in attribute revenue number are respectively 22,000 and 7,000. In the case of Z-Score standardization, the revenue of 42,000 is transferred to Z-Score.

$$\frac{42000 - 22000}{7000} = 2.85$$

3. Decimal Scaling Normalization

- Standardizes the variable values by changing their decimal position.
- We can measure the number of points that the decimal point is passed by the absolute maximum value of the X attribute.
- The value of v of the X attribute is computer-standardized to as:

$$v' = \frac{v}{10^J} \quad \text{where, J is such an integer that maximum.}$$

Example:

CGPA	Formula	CGPA Normalized after Decimal scaling
2	2/10	0.2
3	3/10	0.3

DATA SELECTION AND DATA EXTRACTION

- The main goal of **data selection** is to choose the proper data type, source, and tool that enables researchers to effectively solve research issues.
- This decision typically depends on the discipline and is influenced by the research that has already been done, the body of that research, and the availability of the data sources.

Data Selection Issues

When choosing data, researchers should be conscious of a few things, including:

- Researchers can appropriately respond to the stated research questions when the right type and appropriate data sources are used.
- Appropriate methods for obtaining a representative sample are used.
- The appropriate tools for gathering data are used. It is difficult to separate the choice of data type and source from the tools used to get the data. The type/source of data and the procedures used to collect it should be compatible.

- **Data extraction** is the process of gathering or obtaining many forms of data from numerous sources, many of which may be erratically organized or wholly unstructured.
- Consolidating, processing and refining data enable information to be kept in a centralized area so that it can be altered.
- These venues could be local, online, or a combination of the two.

Data extraction tools:

1. Scrape storm
2. Altair Monarch
3. Klippa
4. NodeXL

Data Extraction and ETL:

1. Extraction
2. Transformation
3. Loading

DATA CURATION:

Data curation is creating, organizing and managing data sets so that people looking for information can access and use them.

Importance of Data Curation:

1. Helps to organize pre-existing data for a corporation
2. Connects professionals in different departments
3. High-quality data typically uses organizational techniques that make it simple to grasp and have fewer errors.
4. Makes data easy to understand
5. Allows for higher cost and time efficiency

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 - **Knowledge Discovery**
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Data Integration Approaches:

1. Tight Coupling- It combines data from many sources into a single physical location using ETL (Extraction, Transformation, and Loading) tools.
2. Loose Coupling- The real source databases are the most efficient place to store facts with loose coupling.
 - This method offers an interface that receives a user query, converts it into a format that the source databases can understand, and immediately transmits the question to the source databases to get the answer.

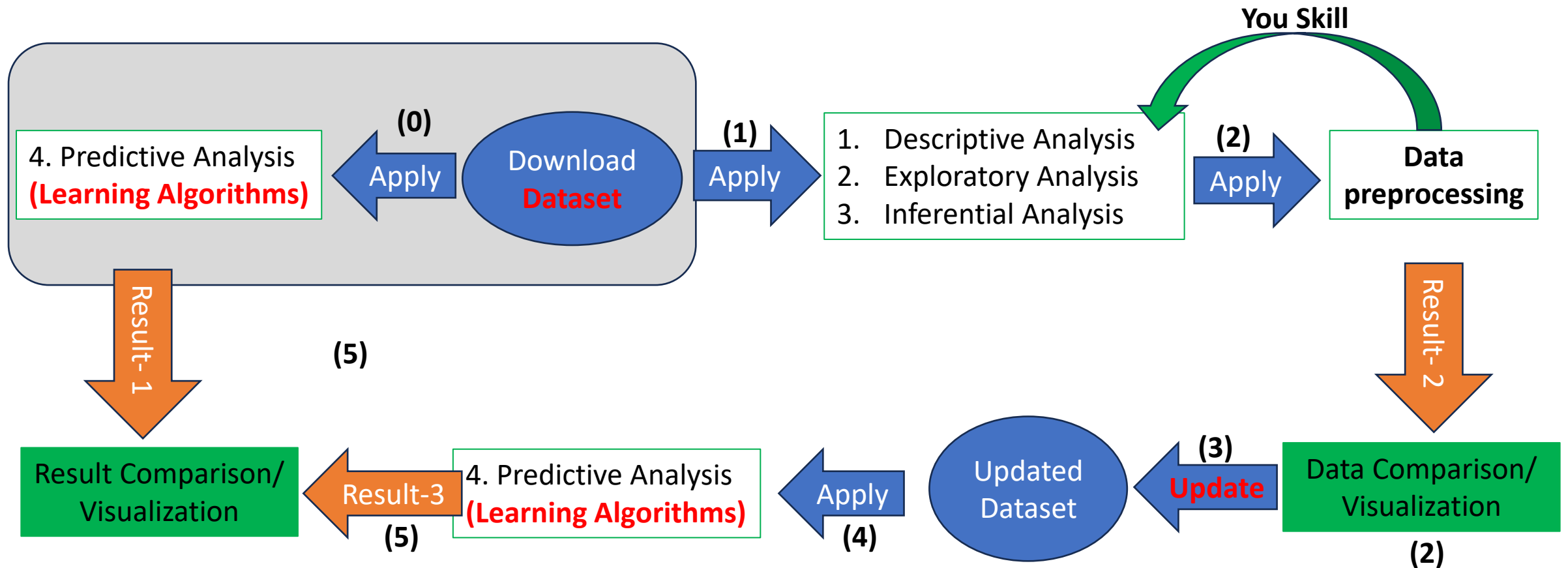
- **Data integration** creates coherent data storage by combining data from several sources.
- Smooth data integration is facilitated by the resolution of semantic heterogeneity, metadata, correlation analysis, tuple duplicate identification, and data conflict detection.
- It is a tactic that combines data from several sources so that consumers may access it in a single, consistent view that displays their status.
- Systems can communicate using flat files, data cubes, or numerous databases.

Data Integration Techniques:

1. Manual Integration
2. Middleware Integration
3. Application-based integration
4. Uniform Access Integration

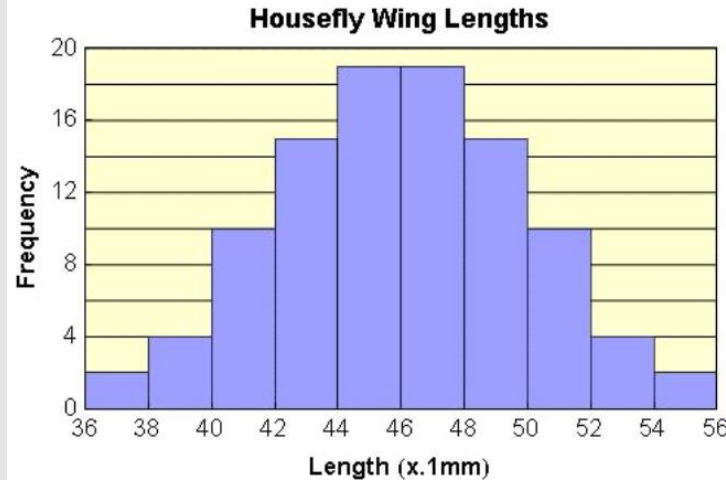
- **Knowledge discovery** in databases is the process of obtaining pertinent knowledge from a body of data (KDD).
- This well-known knowledge discovery method includes several processes, such as
 - data preparation and selection,
 - data cleansing,
 - incorporating prior knowledge about the data sets, and
 - interpreting precise answers from the observed results.
- Marketing, fraud detection, telecommunications, and manufacturing are some of the key KDD application areas.
- In the last ten years, the KDD process has reached its pinnacle.
- Extraction of high-level knowledge from low-level data is the ultimate objective.

Basic idea to build your Projects on Data Science



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Types of HISTOGRAMS:

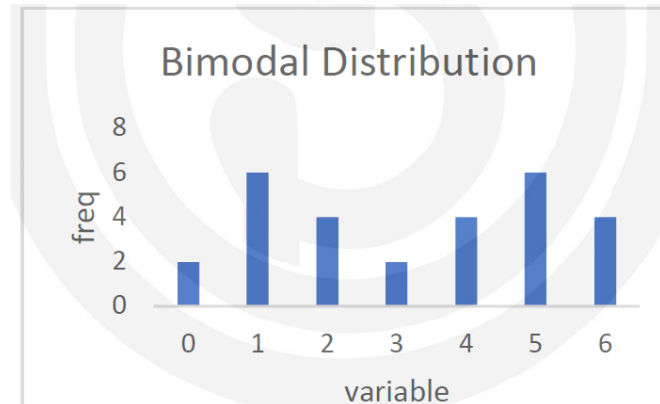


Normal distribution:

- In a normal distribution, the probability that points will occur on each side of the mean is the same.
- This means that points on either side of the mean could occur.

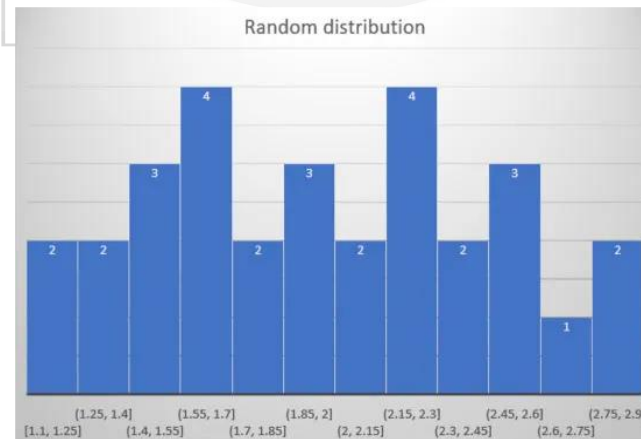
Bimodal Distribution:

- This distribution has two peaks.
- In the case of a bimodal distribution, the data must be segmented before being analysed as normal distributions in their own right.



A random distribution:

- A random distribution is characterised by the absence of any clear pattern and the presence of several peaks.
- When constructing a histogram using a random distribution, it is possible that several distinct data attributes will be blended into one.
- As a result, the data ought to be partitioned and investigated independently.



Histogram:

- A histogram visualises the distribution of data across distinct groups with continuous classes.
- It is represented with set of rectangular bars with widths equal to the class intervals and areas proportional to frequencies in the respective classes.

Constructing a Histogram:

- To construct a histogram, the data is grouped into specific class intervals, or “bins” and plotted along the x-axis.
- These represent the range of the data.
- Then, the rectangles are constructed with their bases along the intervals for each class.
- The height of these rectangles is measured along the y-axis representing the frequency for each class interval.

Use Cases:

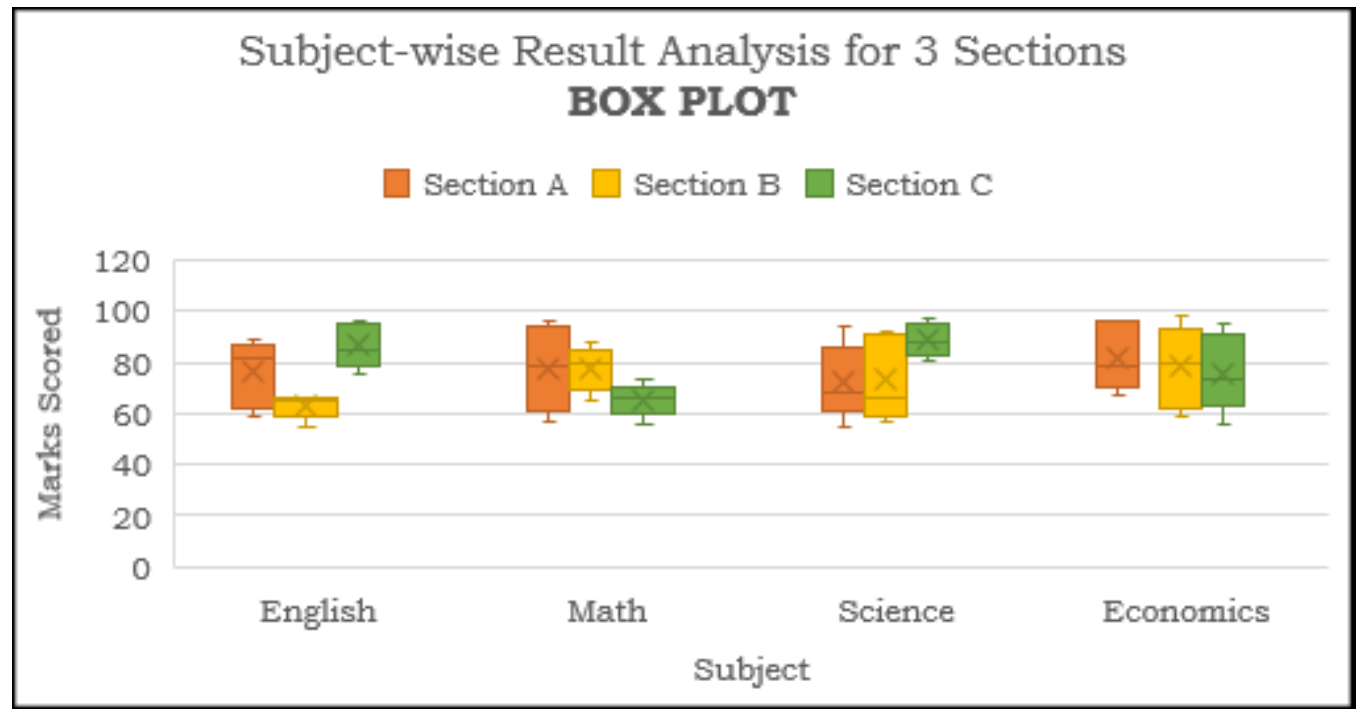
- When it is necessary to illustrate or compare the distribution of specific
- numerical data across several ranges of intervals, histograms can be employed.
- They can aid in visualising the key meanings and patterns associated with a lot of data.
- They may help a business or organization in decision-making process. Some of the use cases of histograms include-
 - ✓ Distribution of salaries in an organisation
 - ✓ Distribution of height in one batch of students of a class, student performance on an exam,
 - ✓ Customers by company size, or the frequency of a product problem.

Best Practices

- **Analyse various data groups:** The best data groupings can be found by creating a variety of histograms.
- **Break down compartments using colour:** The same chart can display a second set of categories by colouring the bars that represent each category.

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- Histograms
- Box plots
- Scatter plots
- Heat map
- Bubble chart
- Bar chart
- Distribution plot
- Pair plot
- Line graph
- Pie chart
- Doughnut chart
- Area chart



- The boxplots clearly shows that
 - ✓ Section B has performed poorly in English, whereas
 - ✓ section C has performed poorly in Maths.
 - ✓ Section A has mostly balanced performance, but the marks of the students are most dispersed.

Boxplot:

- When displaying data distributions using the five essential summary statistics of minimum, first quartile, median, third quartile, and maximum, box-and-whisker plots, also known as boxplots, are widely employed.
- It is a visual depiction of data that aids in determining how widely distributed or how much the data values change.
- These boxplots make it simple to compare the distributions since it makes the centre, spread, and overall range understandable.
- They are utilised for data analysis wherein the graphical representations are used to determine the following:
 1. Shape of Distribution
 2. Central Value
 3. Variability of Data

Constructing a Boxplot:

- The two components of the graphic are described by their
- names: the box, which shows the median value of data along with the first and third quartiles (25 percentile and 75 percentile), and the whiskers, which shows the remaining data.
- The 3rd quartile's difference from the first quartile of data is called the interquartile range.
- The highest and minimum points in the data can also be displayed using the whiskers. The points beyond $1.5 \times$ interquartile range can be identified as suspected outliers.

Use Cases:

- A boxplot is frequently used to demonstrate whether a distribution is skewed and whether the data set contains any potential outliers, or odd observations.
- Boxplots are also very useful for comparing or involving big data sets.
- Examples of box plots include plotting the:
 - ✓ Gas efficiency of vehicles
 - ✓ Time spent reading across readers

Best Practices

- **Cover the points within the box:** This aids the viewer in concentrating on the outliers.
- **Box plot comparisons between categorical dimensions:** Box plots are excellent for quickly comparing dataset distributions.

• UNIT4: DATA VISUALIZATION AND INTERPRETATION

- Histograms
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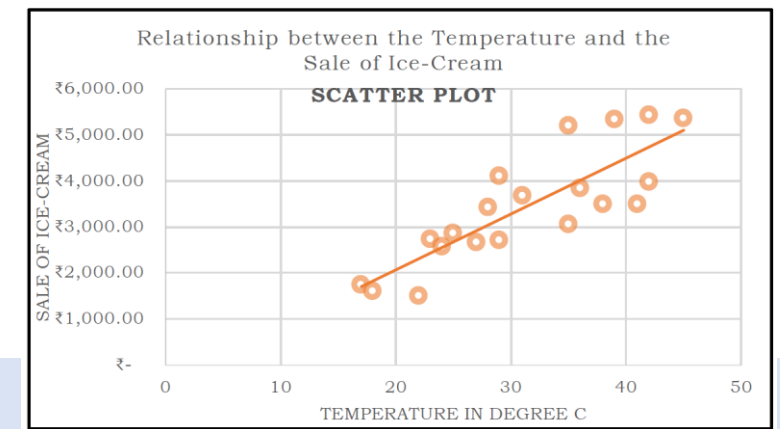
Scatter Plot:

- Scatter plot is the most commonly used chart when observing the relationship between two quantitative variables.
- It works particularly well for quickly identifying possible correlations between different data points.
- The relationship between multiple variables can be efficiently studied using scatter plots,
 - which show whether one variable is a good predictor of another or whether they normally fluctuate independently.

- Note that a linear trendline has been fitted to scatter plot, indicating a positive change in sales of ice-cream with increase in temperature.

Constructing a Scatter Plot:

- Scatter plots are mathematical diagrams or plots that rely on Cartesian coordinates.
- In this type of graph, the categories being compared are represented by the circles on the graph and the numerical volume of the data (indicated by the circle size).



Use Cases:

- Scatter charts are great in scenarios where you want to display both distribution and the relationship between two variables.
- Display the relationship between time-on-platform (How Much Time Do People Spend on Social Media) and churn.
- Display the relationship between salary and years spent at company

Best Practices

- **Analyze clusters to find segments:** Based on your chosen variables, cluster analysis divides up the data points into discrete parts.
- **Employ highlight actions:** You can rapidly identify which points in your scatter plots share characteristics by adding a highlight action, all the while keeping an eye on the rest of the dataset.
- **mark customization:** individual markings Add a simple visual hint to your graph that makes it easy to distinguish between various point groups.

• UNIT4: DATA VISUALIZATION AND INTERPRETATION

- Heat map
- Bubble chart
- Bar chart
- Distribution plot
- Pair plot
- Line graph
- Pie chart
- Doughnut chart
- Area chart

- **Heatmaps** are two-dimensional graphics that show data trends through colour shading.
- Heat maps can be presented in a variety of ways, but they all have one thing in common: **they all make use of colour to convey correlations between data values.**
- The greater the quantity/volume, the deeper the colour (the higher the value, the tighter the dispersion, etc.).
- It improves the ability of existing data visualisations to quickly convey important data insights.

ZONE	MONTH											
	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
NORTH	75	84	61	95	77	82	74	92	58	90	54	83
SOUTH	50	67	89	61	91	77	80	72	82	78	58	63
EAST	62	50	83	95	83	89	72	96	96	81	86	82
WEST	69	73	59	73	57	61	58	60	97	55	81	92

- The distribution of sales is shown in the sample heatmap above, broken down by zone and spanning a 12-month period.
- Like in a typical data table, each cell displays a numeric count, but the count is also accompanied by a colour, with higher counts denoting deeper hues.

Use Cases: Heat Maps are primarily used to better show the enormous amounts of data contained inside a dataset and help guide users to the parts of data visualisations that matter most.

- Average monthly temperatures across the years
- Departments with the highest amount of attrition over time.
- Traffic across a website or a product page.
- Population density/spread in a geographical location.

Best Practices

- **Select the proper colour scheme:** This style of chart relies heavily on colour, therefore it's important to pick a colour scheme that complements the data.
- **Specify a legend:** As a related point, a heatmap must typically contain a legend describing how the colours correspond to numerical values.

• UNIT4: DATA VISUALIZATION AND INTERPRETATION

- Heat map
 - Bubble chart
 - Bar chart
 - Distribution plot
 - Pair plot
 - Line graph
 - Pie chart
 - Doughnut chart
 - Area chart
-
- **Bubble** diagrams are used to show the relationships between different variables.
 - They are frequently used to represent data points in three dimensions, specifically when the bubble size, y-axis, and x-axis are all present.
 - Using location and size, bubble charts demonstrate relationships between data points.

- The three variables in this example are sales, profits, and the number of units sold.
- Therefore, all three variables and their relationship can be displayed simultaneously using a bubble chart.

Constructing a Bubble Chart:

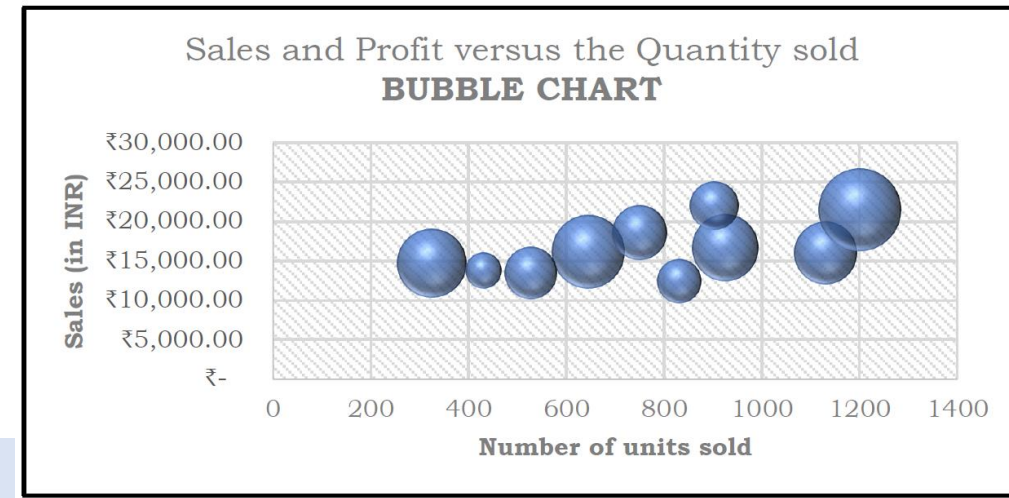
- For each observation of a pair of numerical variables (A, B), a bubble or disc is drawn and placed in a Cartesian coordinate system horizontally according to the value of variable A and vertically according to the value of variable B.
- The area of the bubble serves as a representation for a third numerical variable (C).
- Using various colours in various bubbles, we may even add a fourth dataset (D: numerical or categorical).

Use Cases:

- Usually, the positioning and ratios of the size of the bubbles/circles on this chart are used to compare and show correlations between variables.
- Additionally, it is utilised to spot trends and patterns in data.
 - ✓ AdWords' analysis: CPC vs Conversions vs share of total conversions
 - ✓ Relationship between life expectancy, GD per capita and population size

Best Practices:

- **Add colour:** A bubble chart can gain extra depth by using colour.
- Set bubble size in appropriate proportion.
- **Overlay bubbles on maps:** From bubbles, a viewer can immediately determine the relative concentration of data.

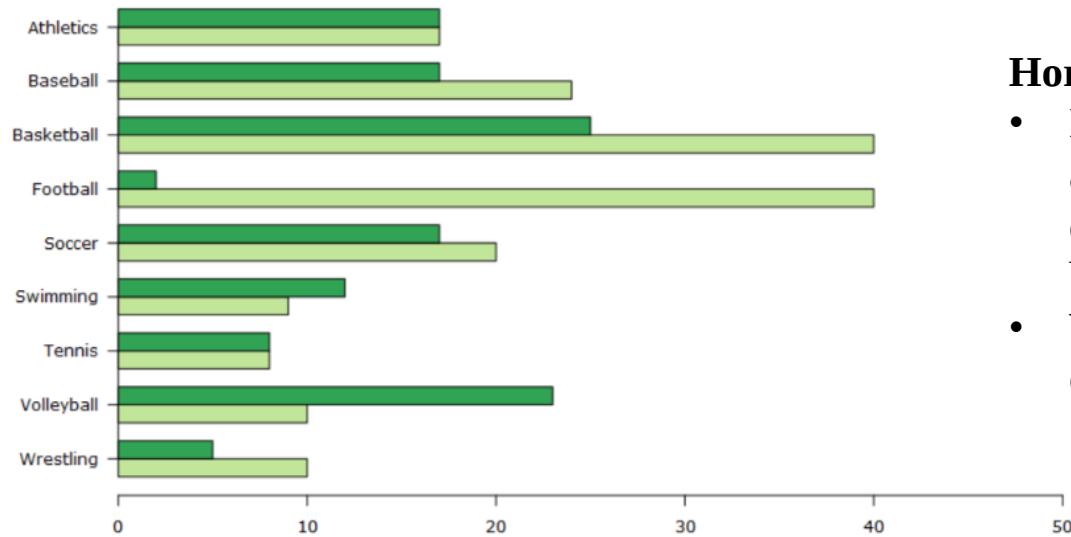


• UNIT4: DATA VISUALIZATION AND INTERPRETATION

• Bar chart

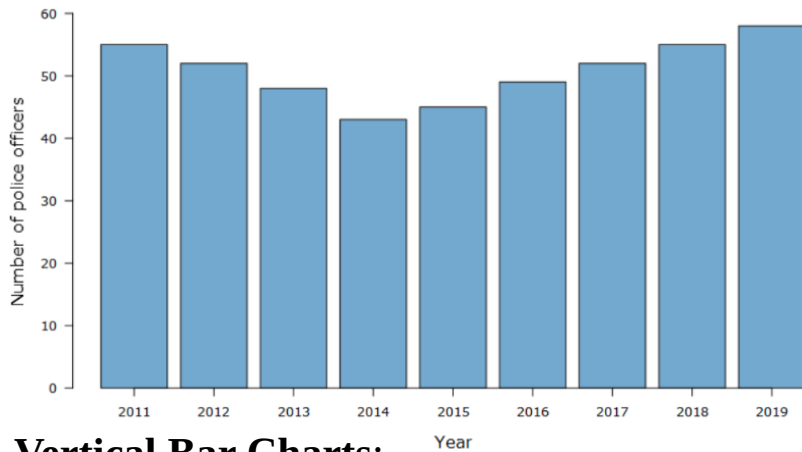
- Distribution plot
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Types of Bar chart: Bar Charts are mainly classified into two types:



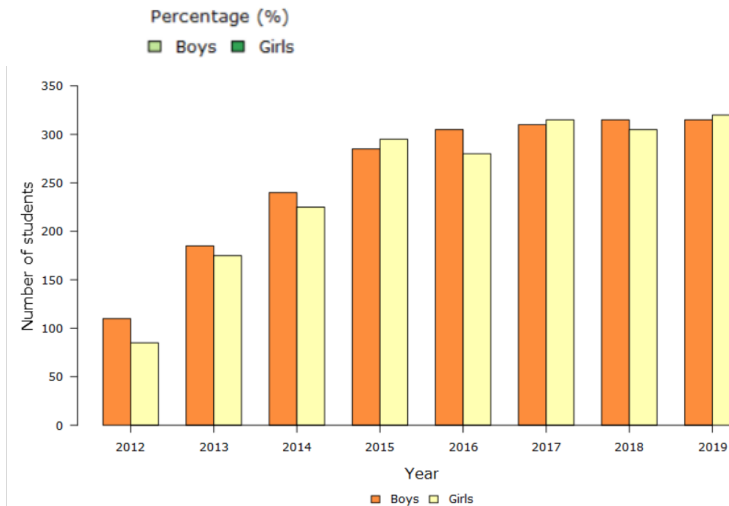
Horizontal Bar Charts:

- Horizontal bar charts are the type of graph that are used when the data being analysed is to be depicted on paper in the form of horizontal bars with their respective measures.
- When using a chart of this type, the categories of the data are indicated on the y-axis.



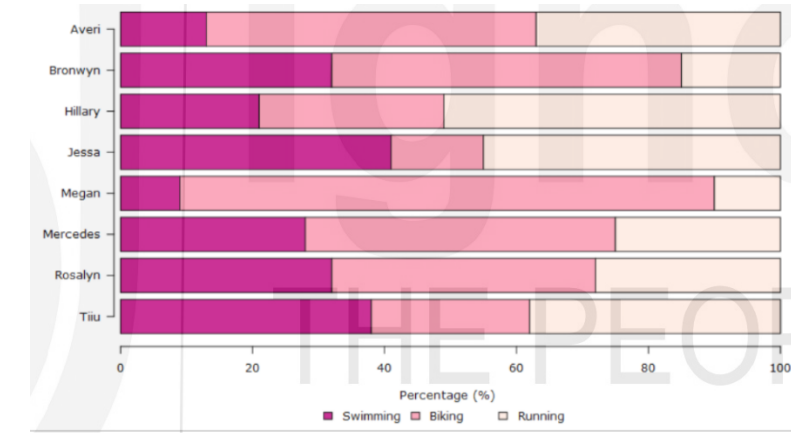
Vertical Bar Charts:

- A vertical bar chart displays vertical bars on graph (chart) paper.
- These rectangular bars in a vertical orientation represent the measurement of the data.
- The quantities of the variables that are written along the x-axis are represented by these rectangular bars.



Grouped Bar Charts:

- The grouped bar graph is also referred to as the clustered bar graph (graph).
- It is valuable for at least two separate types of data.



Stacked Bar Charts:

- The composite bar chart is also referred to as the stacked bar chart.
- It illustrates how the overall bar chart has been broken down into its component pieces.

Bar Chart:

- A bar chart is a graphical depiction of numerical data that uses rectangles (or bars) with equal widths and varied heights.
- In the field of statistics, bar charts are one of the methods for handling data.

Constructing a Bar Chart:

- The x-axis corresponds to the horizontal line, and the y-axis corresponds to the vertical line.
- The y-axis represents frequency in this graph.
- Write the names of the data items whose values are to be noted along the x-axis that is horizontal.
- Along the horizontal axis, choose the uniform width of bars and the uniform gap between the bars.
- Pick an appropriate scale to go along the y-axis that runs vertically so that we can figure out how high the bars should be based on the values that are presented.
- Determine the heights of the bars using the scale we selected, then draw the bars using that information.

Use cases:

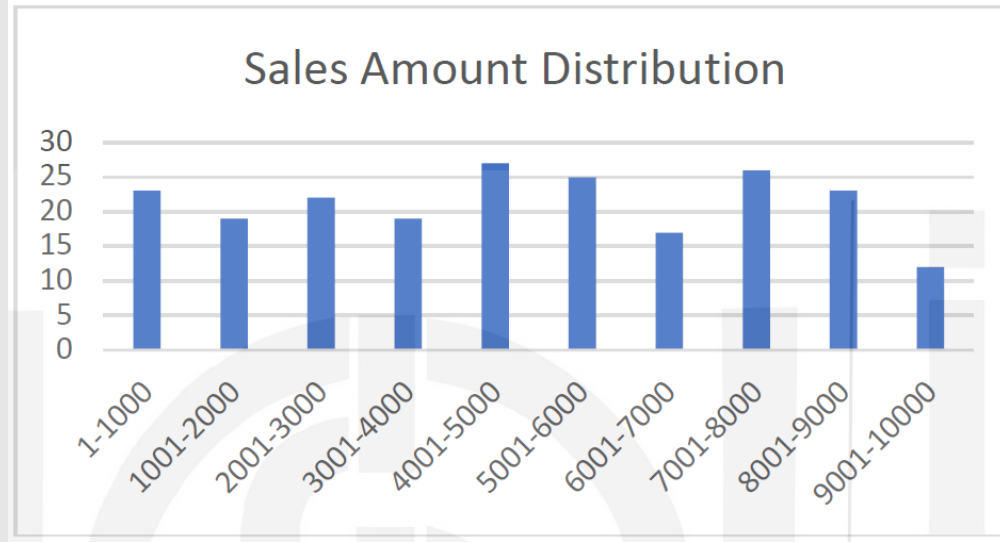
- Bar charts are typically employed to display quantitative data.
- The following is a list of some of the applications of the bar chart-
 - ✓ In order to clearly illustrate the relationships between various variables, bar charts are typically utilised. When presented in a pictorial format, the parameters can be more quickly and easily envisioned by the user.
 - ✓ Bar charts are the quickest and easiest way to display extensive amounts of data while saving time. It is utilised for studying trends over extended amounts of time.

Best Practices:

- Use a common zero valued baseline
- Maintain rectangular forms for your bars
- Consider the ordering of category level and use colour wisely.

• UNIT4: DATA VISUALIZATION AND INTERPRETATION

- **Distribution plot**
- Pair plot
- Line graph
- Pie chart
- Doughnut chart
- Area chart



Constructing a Distribution Plot:

- We must utilise one or two dimensions, together with one measure, in a distribution plot.
- We will get a single line visualisation if we only use one dimension.
- If we use two dimensions, each value of the outer, second dimension will produce a separate line.

Distribution Plot:

- Visually assessing the distribution of sample data, distribution charts do this by contrasting the actual distribution of the data with the theoretical values expected from a certain distribution.
- In addition to more traditional hypothesis tests, distribution plots can be used to establish whether the data from the sample follows a particular distribution.
- The distribution plot is useful for analysing the relationship between the range of a set of numerical data and its distribution.
- The values of the data are represented as points along an axis.

Use Cases:

- Distribution of a data set shows the frequency of occurrence of each possible outcome of a repeatable event observed many times.
- For instance:
 - ✓ Height of a population.
 - ✓ Income distribution in an economy
 - ✓ Test scores listed by percentile.

Best Practices:

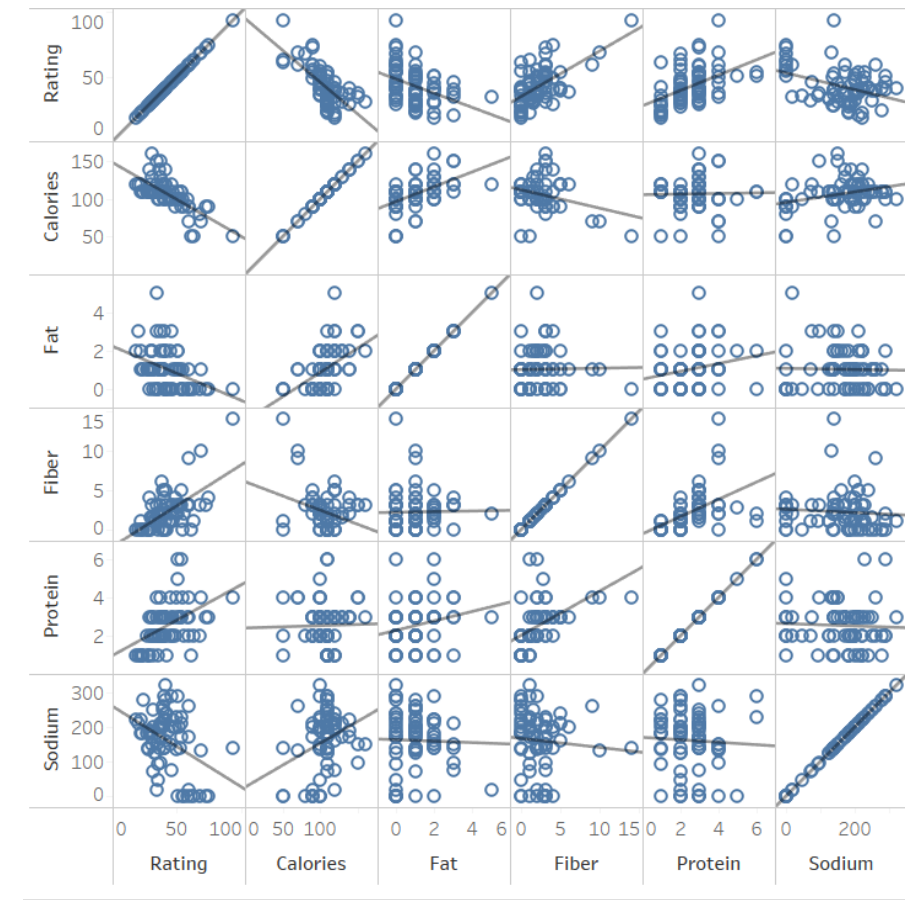
- It is advisable to have equal class widths.
- The class intervals should be mutually exclusive and non-overlapping.
- Open-ended classes at the lower and upper limits (e.g., <10 , >100) should be avoided.

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The pair plot can be interpreted as follows:

- Along the boxes of the diagonal, the variable names are displayed.
- A scatterplot of the correlation between each pairwise combination of factors is shown in each of the remaining boxes.
- For instance, a scatterplot of the values for rating and sodium can be seen in the matrix's box in the top right corner.
- A scatterplot of values for rating, that is positively connected with rating, and so forth may be seen in the box in the upper left corner.
- We can see the association between each pair of variables in our dataset from this single visualisation.
- For instance, calories and rating appear to have a negative link but protein and fat appear to be unrelated.



- The **pairs plot** is an extension of the histogram and the scatter plot, which are both fundamental figures.
- The scatter plots on the upper and lower triangles show the relationship (or lack thereof) between two variables.
- The histogram along the diagonal gives us the ability to see the distribution of a single variable, while the scatter plots on the upper and lower triangles show the relationship (or lack thereof) between two variables.
- A pair plot can be utilised to gain an understanding of the optimum collection of characteristics to describe a relationship between two variables or to create clusters that are the most distinct from one another.
- Additionally, it is helpful to construct some straightforward classification models by drawing some straightforward lines or making linear separations in our data set.

Constructing a Pair Plot:

- If you have m attributes in your dataset, it creates a figure with $m \times m$ subplots.
- Each attribute's univariate histograms (distributions) make up the main-diagonal subplots.
- For a non-diagonal subplot, assume a position (i, j) .
- The dataset's samples are all plotted using a coordinate system with the characteristics i and j as the axes.
- In other words, it projects the dataset on these two attributes only.
- This is particularly interesting to visually inspect how the samples are spread with respect to these two attributes ONLY.
- The "shape" of the spread can give you valuable insight on the relation between the two attributes.

Use Cases:

- A pairs plot allows us to see both distribution of single variables and relationships between two variables.
- It helps to identify the most distinct clusters or the optimum combination of attributes to describe the relationship between two variables.
 - ✓ By creating some straightforward linear separations or basic lines in our data set, it also helps to create some straightforward classification models.
 - ✓ Analysing socio-economic data of a population.

Best Practices:

- Use a different colour palette.
- For each colour level, use a different marker.

- **UNIT4: DATA VISUALIZATION AND INTERPRETATION**

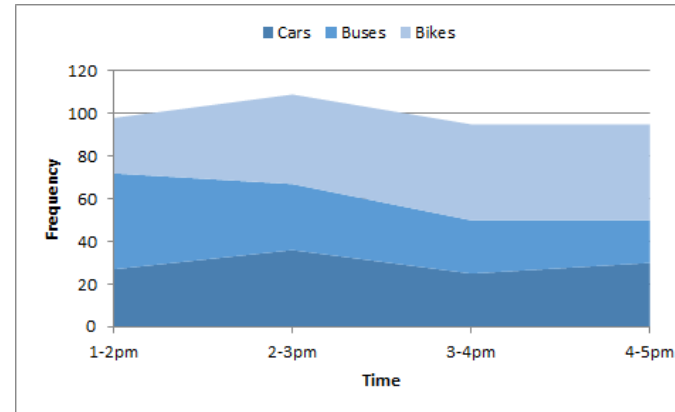
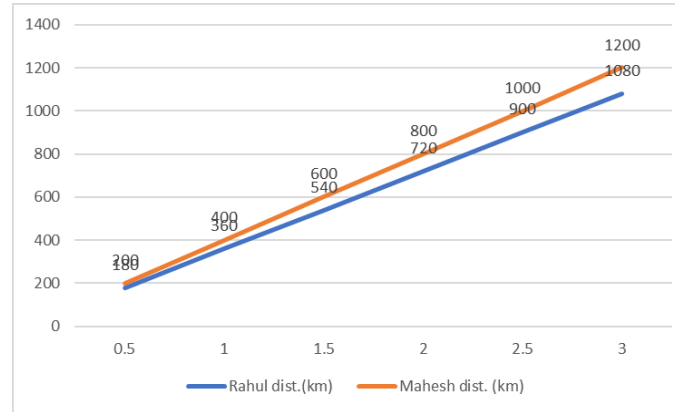
- Distribution plot
- Pair plot
- Line graph
- Pie chart
- Doughnut chart
- Area chart

Line Graph:

- A graph that depicts change over time by means of points and lines is known as a line graph, line chart, or line plot.
- It is a graph that shows a line connecting a lot of points or a line that shows how the points relate to one another.

Types of Line Graphs:

1. **Simple Line Graph:** Only a single line is plotted on the graph.



2. **Multiple Line Graph:**

- The same set of axes is used to plot several lines.
- An excellent way to compare similar objects over the same time period is via a multiple line graph.

3. **Compound Line Graph:**

- Whenever one piece of information may be broken down into two or more distinct pieces of data.
- A compound line graph is the name given to this particular kind of line graph.
- To illustrate each component that makes up the whole, lines are drawn.
- The line at the top displays the total, while the line below displays a portion of the total.
- The size of each component can be determined by the distance that separates every pair of lines.

It is strongly advised to use a line graph rather than a bar graph when working with data that only has slight fluctuations.

Constructing a line graph:

- When we have finished creating the data tables, we will then use those tables to build the linear graphs.
- These graphs are constructed by plotting a succession of points, which are then connected together with straight lines to offer a straightforward method for analysing data gathered over a period of time.
- It provides a very good visual format of the outcome data that was gathered over the course of time.

Use cases:

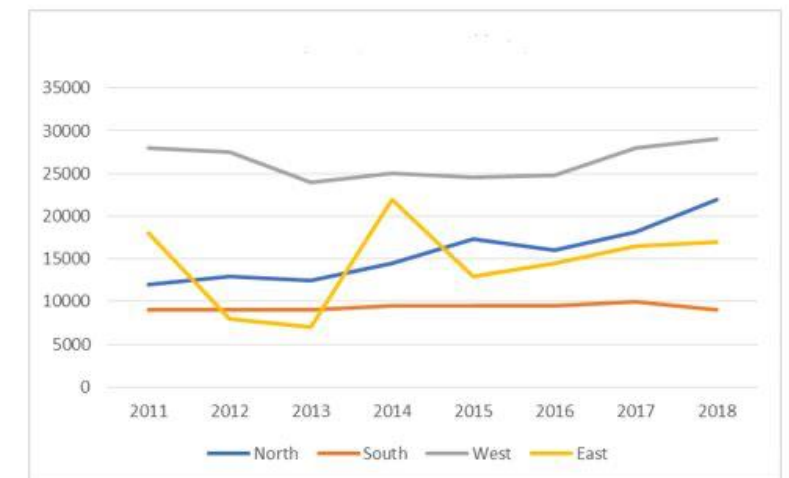
- Tracking changes over both short and long time periods is an important application of line graphs.
- Additionally, it is utilised to compare changes over the same time period for various groups.
- Anytime there are little changes, using a line graph rather than a bar graph is always preferable.

Best Practices

- Only connecting adjacent values along an interval scale should be done with lines.
- In order to provide correct insights, intervals should be of comparable size.
- Select a baseline that makes sense for your set of data; a zero baseline might not adequately capture changes in the data.
- Line graphs are only helpful for comparing data sets if the axes have the same scales.

Example:

Sales	2011	2012	2013	2014	2015	2016	2017	2018
North	12000	13000	12500	14500	17300	16000	18200	22000
South	9000	9000	9000	9500	9500	9500	10000	9000
West	28000	27500	24000	25000	24500	24750	28000	29000
East	18000	8000	7000	22000	13000	14500	16500	17000



• UNIT4: DATA VISUALIZATION AND INTERPRETATION

- Distribution plot
- Pair plot
- Line graph
- Pie chart
- Doughnut chart
- Area chart

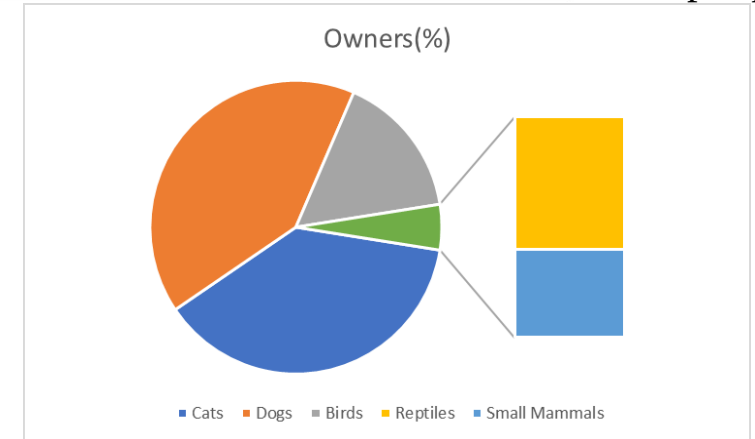
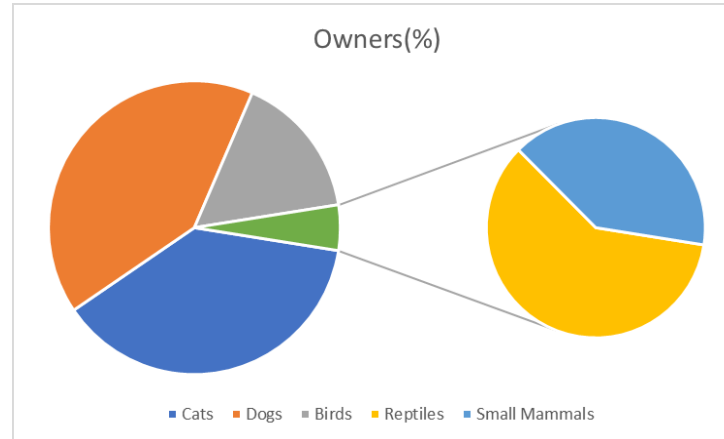
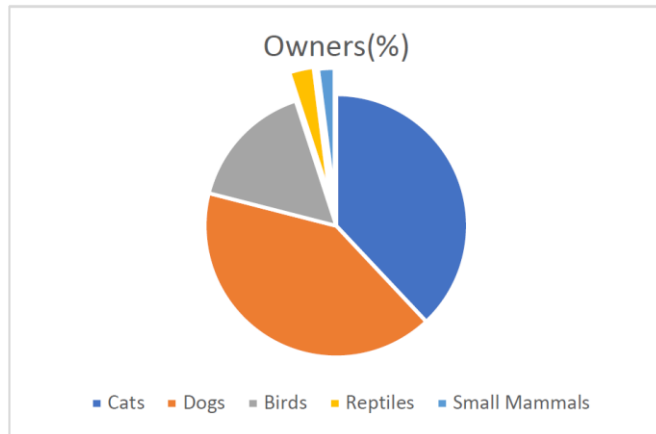
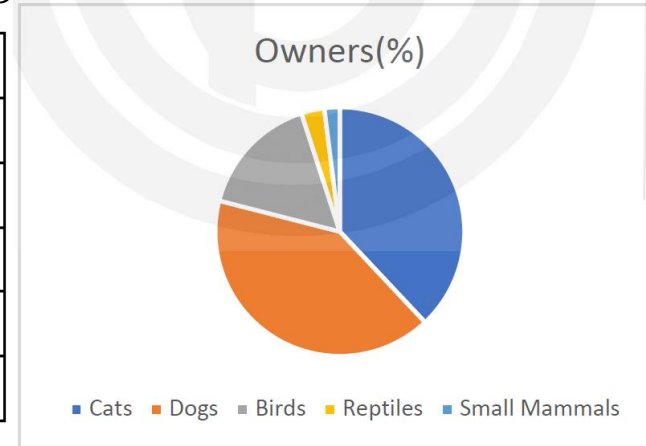
Types of a Pie chart:

There are mainly two types of pie charts one is 2D pie chart and another is 3D pie chart. This can be further classified into following categories:

Pets	Owners (%)
Cats	38
Dogs	41
Birds	16
Reptiles	3
Small Mammals	2

1. Simple Pie Chart:

- The most fundamental kind of pie chart is referred to simply as a pie chart and is known as a simple pie chart.



2. Exploded Pie Chart:

- To create an exploding pie chart, we must first separate the pie from the chart itself, as opposed to merging the two elements together.

3. Pie of Pie:

- A pie chart that is generated from an already existing pie chart is referred to as a "pie of pie".

4. Bar of Pie:

- The expansion that was developed from the already existing pie chart was a bar graph rather than a pie of pie, despite the fact that both serve comparable objectives.

Constructing a Pie chart:

- “The total value of the pie is always 100%”
- To work out with the percentage for a pie chart, follow the steps given below:
 - ✓ Categorize the data
 - ✓ Calculate the total
 - ✓ Divide the categories
 - ✓ Convert into percentages
 - ✓ Finally, calculate the degrees

Therefore, the pie chart formula is given as (Given Data/Total value of Data) × 360°

Best Practices

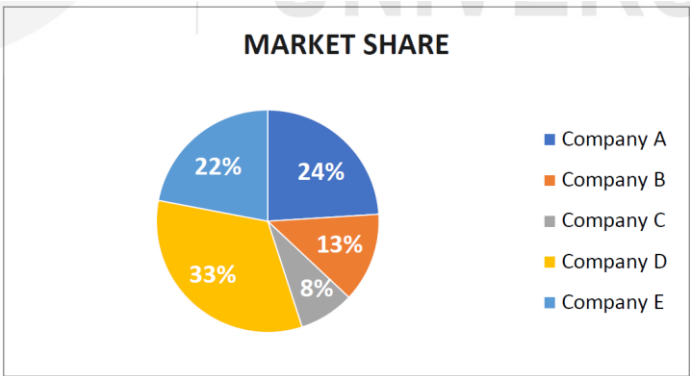
- **Fewer pie wedges are preferred:** The observer may struggle to interpret the chart’s significance if there are too many proportions to compare. Similar to this, keep the overall number of pie charts on dashboards to a minimum.
- **Overlay pies on maps:** Pie charts can be used to further deconstruct geographic tendencies in your data and produce an engaging display.

Use cases:

- If you want your audience to get a general idea of the part-to-whole relationship in your data, and comparing the exact sizes of the slices is not as critical to you, then you should use pie charts.
- And indicate that a certain portion of the whole is disproportionately small or large.
- ✓ Voting preference by age group
- ✓ Market share of cloud providers

Example:

COMPANY	MARKET SHARE
Company A	24%
Company B	13%
Company C	8%
Company D	33%
Company E	22%

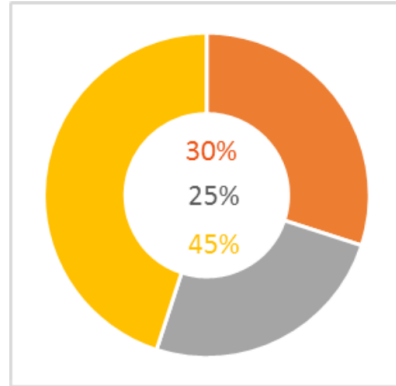


• UNIT4: DATA VISUALIZATION AND INTERPRETATION

- Distribution plot
- Pair plot
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- Pie chart
- Doughnut chart
- Area chart

Example:

Project Status	
Completed	30%
Work in progress	25%
Incomplete	45%



Constructing a Doughnut chart:

- A doughnut chart, like a pie chart, illustrates the relationship of individual components to the whole, but unlike a pie chart, it can display more than one data series at the same time.
- A specific kind of pie chart called a doughnut chart is used to show the percentages of categorical data.
- There are two types of doughnut chart one is normal doughnut chart and another is exploded doughnut chart.
- Exploding doughnut charts, much like exploded pie charts, highlight the contribution of each value to a total while emphasising individual values.
- However, unlike exploded pie charts, exploded doughnut charts can include more than one data series.

Doughnut chart:

- Pie charts have been superseded by a more user-friendly alternative called a doughnut chart, which makes reading pie charts much simpler.
- It is recognised that these charts express the relationship of 'part-to-whole,' which is when all of the parts represent one hundred percent when collected together.

Use cases:

- Doughnut charts are good to use when comparing sets of data.
- By using the size of each component to reflect the percentage of each category, they are used to display the proportions of categorical data.
- A string field and a count of features, number, rate/ratio, or field are used to make a doughnut chart.
 - ✓ Android OS market share
 - ✓ Monthly sales by channel

Best Practices

- Stick to five slices or less because thinner and long-tail slices become unreadable and uncomparable.
- Use this chart to display one point in time with the help of the filter legend.
- Well-formatted and informative labels are essential because the information conveyed by circular shapes alone is not enough and is imprecise.
- It is a good practice to sort the slices to make it more clear for comparison.

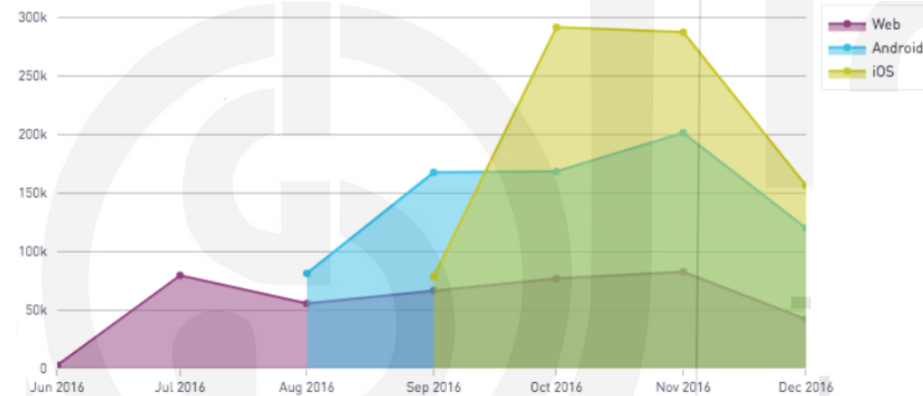
• UNIT4: DATA VISUALIZATION AND INTERPRETATION

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Types of Area Chart:

Months (2016)	Web	Android	IOS
June	0	-	
July	70k	-	
Aug	55k	80k	
Sep	60k	165k	80k
Oct	70k	165k	295k
Nov	80k	200k	290k
Dec	40k	125k	155k

Overlapping Area Chart

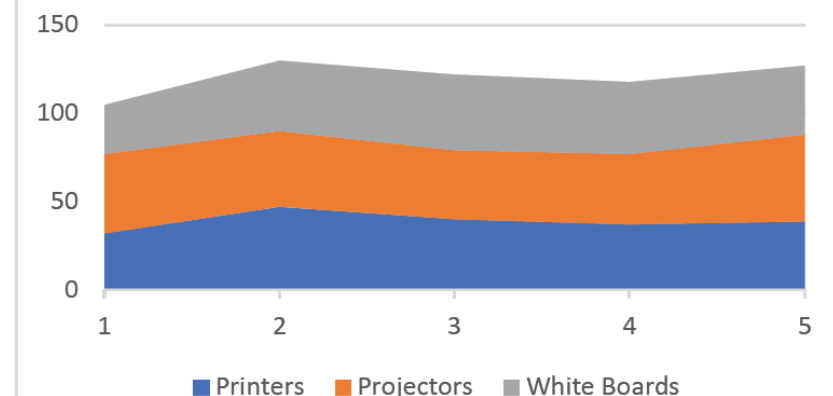


Overlapping area chart:

An overlapping area chart results if we wish to look at how the values of the various groups compare to one another.

A	B	C	D
	Printers	Projectors	White Boards
2017	32	45	28
2018	47	43	40
2019	40	39	43
2020	37	40	41
2021	39	49	39

Stacked Area chart



Stacked area chart:

The stacked area chart is what is often meant to be conveyed when the phrase "area chart" is used in general conversation.

Area Chart :

- An area chart, a hybrid of a line and bar chart, shows the relationship between the numerical values of one or more groups and the development of a second variable, most often the passage of time.

Use Cases:

- In most cases, many lines are drawn on an area chart in order to create a comparison between different groups (also known as series) or to illustrate how a whole is broken down into its component pieces.
- This results in two distinct forms of area charts, one for each possible application of the chart.
- ✓ **Magnitude of a single quantitative variable's trend** - An increase in a public company's revenue reserves, programme enrollment from a qualified subgroup by year, and trends in mortality rates over time by primary causes of death are just a few examples.
- ✓ **Comparison of the contributions made by different category members (or group)**- the variation in staff sizes among departments, or support tickets opened for various problems.
- ✓ Birth and death rates over time for a region, the magnitudes of cost vs. revenue for a business, the magnitudes of export vs. import over time for a country

Best Practices:

- To appropriately portray the proportionate difference in the data, start the y-axis at 0.
- To boost readability, choose translucent, contrasting colours.
- Keep highly variable data at the top of the chart and low variable data at the bottom during stacking.
- If you need to show how each value over time contributes to a total, use a stacked area chart.
- However, it is recommended to utilise 100% stacked area charts if you need to demonstrate a part to whole relationship in a situation where the cumulative total is unimportant.